

Development of a wearable device for monitoring the activity of elderly people with dementia: first insights

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Abstract. The world has been witnessing a demographic shift with an ageing population, leading to an increase in the prevalence of neurodegenerative diseases. Among all, dementias represent one of the major causes of dependency and morbidity among the elderly, causing changes in the behavior, memory, and the ability to perform daily tasks. The provision of care for the elderly faces significant challenges, with caregivers needing to develop physical and psychological capacities to perform their duties. This work presents the first steps in the development of a wearable device (wristband) for real-time activity monitoring, promoting adequate and minimally invasive care for elderly people with dementia. The wristband methodology to be developed is divided into two parts, the functionalities methodology and the design methodology, which are responsible for the functional and non-functional requirements.

Keywords: Wearable Device · Elderly People · Dementia · Design

1 Introduction

The United Nations notes that the global population is rapidly ageing, with all countries seeing a rise in elderly populations. This demographic shift has led to a sharp increase in dementia cases, affecting over 55 million people [1]. Caregivers play a fundamental role in the care of elderly with dementia, helping them to carry out daily activities, and their role is associated with great responsibility and physical and psychological demands [2]. Therefore, recognizing and understanding the specific needs faced by this target audience is essential for designing tailored interventions and systems that contribute to a better quality of life both for elderly and their caregivers [3]. The project where this work is included "Platform for Routine Measurement and Monitoring of Activity in Elderly People with Dementia" aims to develop and validate a new system for location and activity monitoring, promoting active, healthy, and inclusive ageing. This new system will feature two wearable data acquisition devices, the HowMi [4] (in-ear) and a wristband – which is the aim of the work presented in this paper. These devices will acquire biometric data such as body temperature, heart rate, blood oxygen saturation level, but also daily activity data (activity index, number of steps and distance traveled, fall detection, sleep rhythm analysis) and indoor and outdoor location. The data will be sent to a server, where it will be managed and monitored. The information will be made available on a web portal for institutional professionals, caregivers, and family members, helping them to decide on the most appropriate type of care for the

elderly person’s situation. This paper is organized in 4 sections, Section 2 ”Related Work” describes the current literature framework in the area of wearable devices in the context of monitoring the activity parameters of elderly people with dementia, Section 3 ”Methodology” describes the approach used for achieve the setting goals, and Section 4 ”Future Work” describes the next steps that will be taken to continue the work according to the methodologies defined.

2 Related work

The evolution of wearable devices for monitoring daily activities has garnered significant importance in recent years for promoting safe, non-invasive, and cost-effective monitoring through intuitive use. Several studies have focused on the incorporation of wearable devices to monitor health metrics, behavioral and sleep patterns, activity levels and real-time tracking in vulnerable population. This section provides an overview of recent and notable contributions in this field. One of the studies in this domain explored the use of the WHOOP biosensor device for monitoring sleep and heart rate variability in individuals at risk for Alzheimer’s disease [5]. Their findings demonstrated that wearable devices can allow caregivers or clinician researchers, an early detection and continuous monitoring, thereby facilitating timely and individualized interventions to manage and potentially delay the progression of cognitive decline. In addition, another study used an accelerometer wearable device for examine whether Parkinson’s disease (PD) intensity, influences the relationship between functional connectivity and cognitive performance [6]. Their work revealed that performing moderate physical activity can improve cognitive functionality in individuals with PD, such as visuospatial function, memory, and executive function. Recent advancements have also been made in the field of artificial intelligence and the development of multi-modal sensors integrated in wearable devices for measuring a range of physiological and environmental parameters. Given this, a recent study introduced a multi-sensor wearable device to detect behavioral symptoms of dementia, including motor agitation, verbal, and physical aggression [7]. Their study revealed that using a device with personalized sensors can, not only, enhance the performance of generic models in classifying agitated behaviors but understanding the complex and progression of the pathology. Overall, the existing literature underscores the promising potential of wearable devices in improving the care of elderly individuals with dementia or prevent who are heading towards it. Despite the innovative advancements, there are still substantial challenges in terms of long-term usability since certain technologies need improvement once is given the specific needs of elderly individuals with dementia [3]. Another factor in consideration is that caregivers play a fundamental paper on the daily lives of these individuals. It is therefore of the utmost importance the adequately prepare and integration in the assessment and use of these wearable devices, enhancing the relationship between the caregiver and the patient [8]. The continuous collaboration among researches, healthcare providers, technology and design developers can ensure the development of cost-effective wearable devices by reducing the economic burden of dementia care.

3 Methodology

The structure of this work, particularly the design of the wristband, comprises two main sections: the functionalities methodology and the design methodology, as shown in Figure 1.

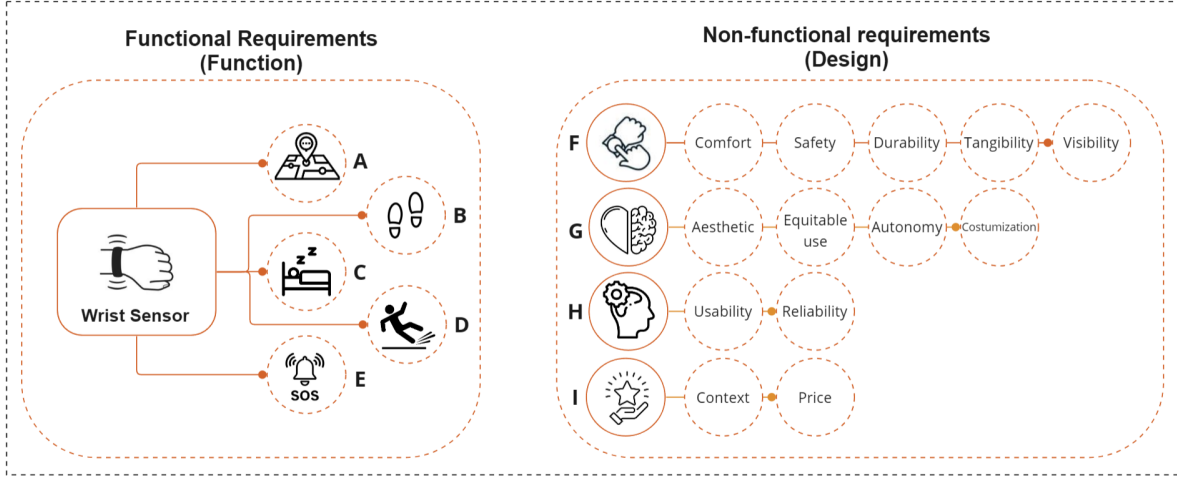


Fig. 1: Functional and non-functional requirements of the wristband.

These are outlined to establish the fundamental functional and non-functional requirements for the design of the wristband wearable device. In terms of functionality, the device must be able to communicate via ISM (Industrial, Scientific and Medical) Radio Frequency with the system's components. This communication will be done using the BLE (Bluetooth Low Energy) protocol in indoor environments and using the LoRa (Long Range) protocol in outdoor environments. Regarding the location requirement (A), indoor location will be given by connecting the wristband with BLE beacon devices and outdoor location will be given by a GPS/GNSS (Global Positioning System/Global Navigation Satellite Systems) module. This system will provide information to family members and caregivers about the patient's real-time location in cases of spatial memory loss episodes. It will have inertial sensors for sensory fusion and consequent analysis of daily activity patterns such as activity index, number of steps and sleep rhythm analysis (B,C). It will also include a fall detection system (D) and an emergency alert system (E), which will inform caregivers for a quick response to the necessary care, as this is a highly vulnerable group in terms of physical and cognitive abilities. On the other hand, non-functional requirements provide guidelines that facilitate the adoption of the wristband. In view of this, four domains of requirements have been defined: the physical domain (F) focuses on the direct relationship between the user and the product. The emotional domain (G) establishes considerations ethical and visual issues, and the cognitive domain (H) ensures the complete usability of the wristband according to the abilities of the elderly. Finally, the requirements of the implementation success domain (I) focus on assisting in the integration of the wristband for the problem at hand.

4 Future Work

Further steps will follow the methodology being applied, starting by selecting the best sensors required for the wristband, including inertial sensors and GPS. We will then analyze and test the communication protocols (Bluetooth Low Energy and LoRa (Long Range)) and the integration with the HowMI device [4], internal location devices and the data server. Additionally, we will develop AI algorithms to analyze the collected data, enabling advanced functionalities such as localization tracking, routine monitoring, sleep pattern analysis, and anomaly detection. Finally, in terms of design, we aim to start ergonomic studies of the wrists and prepare a questionnaire for caregivers in order to obtain feedback on the idealized design of the wristband.

Acknowledgments

This project was funded through the Foundation for Science and Technology (FCT) under the projects UIDB/05549/2020 (DOI: 10.54499/UIDB/05549/2020), UIDP/05549/2020 (DOI: 10.54499/UIDP/05549/2020), LASI-LA/P/0104/2020, and CEECINST/00039/2021. This work was also funded by the Innovation Pact HfFP – Health From Portugal, co-funded from the “Mobilizing Agendas for Business Innovation” of the “Next Generation EU” program of Component 5 of the Recovery and Resilience Plan (RRP), concerning “Capitalization and Business Innovation”, under the Regulation of the Incentive System “Agendas for Business Innovation”.

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